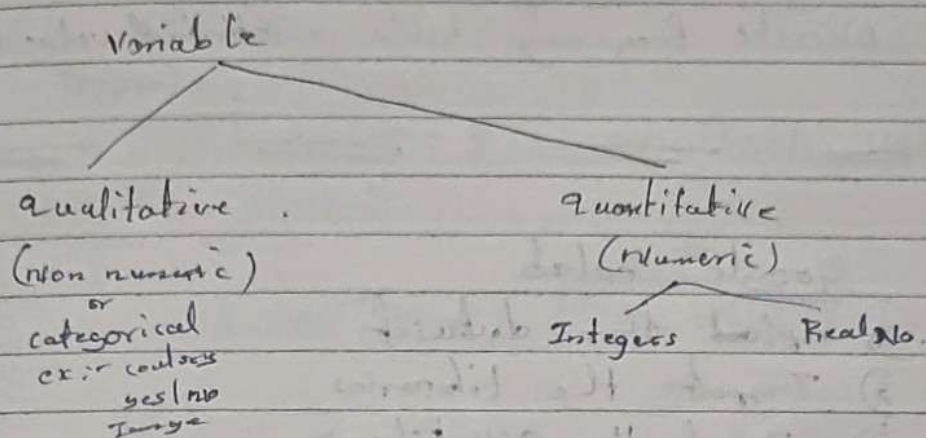


Meaning less qty

- Quantitative data $\rightarrow$ Numeric
- Qualitative data $\rightarrow$ non numeric
- Chronological $\rightarrow$ Time



scales of measurement

Virtual scale.

Properties

- • identification
- • magnitude
- • Equal Intervals
- • True zero (min values)

1) Nominal scale } Qualitative data

- identity

2) ordinal scale } identification - magnitude

- order wise, ascending or descending

3) Interval scale } it doesn't have ~~no~~ no true zero

both qualitative & quantitative

4) Ratio scale (:) it passes true zero & all properties

Quantitative

$w = \text{width}$
 $h = \text{height}$
 $\text{area} = w \times h$
 $\text{volume} = w \times h \times \text{depth}$

1. $\text{area} = \text{length} \times \text{width}$
 2. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

3. $\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$
 4. $\text{volume} = \frac{1}{3} \times \text{base} \times \text{height}$

5. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 6. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

7. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 8. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

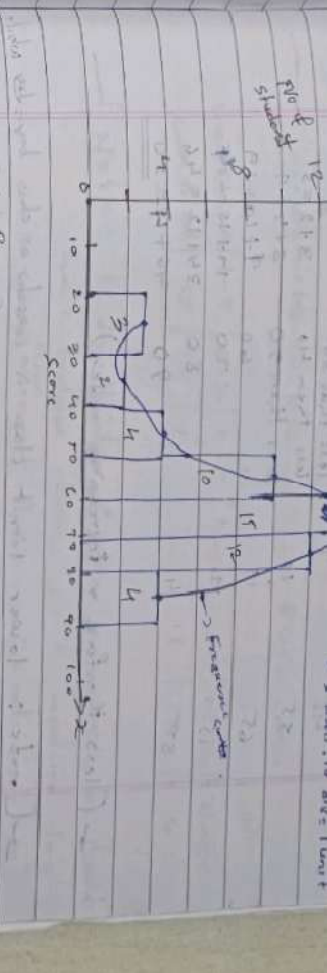
9. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 10. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

11. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 12. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

13. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 14. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

15. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 16. $\text{volume} = \text{length} \times \text{width} \times \text{height}$

17. $\text{area} = \text{length} \times \text{width} \times \text{height}$
 18. $\text{volume} = \text{length} \times \text{width} \times \text{height}$



Draw frequency curve
 Plot class frequency against class mid point $= \frac{L_b + U_b}{2}$

draw relative frequency histogram / relative frequency curve

class	F	Relative Frequency
0-30	3	$\frac{3}{50} = 0.06$
30-40	4	$\frac{4}{50} = 0.08$
40-50	10	$\frac{10}{50} = 0.20$
50-60	15	$\frac{15}{50} = 0.30$
60-70	10	$\frac{10}{50} = 0.20$
70-80	4	$\frac{4}{50} = 0.08$
80-90	1	$\frac{1}{50} = 0.02$
90-100	1	$\frac{1}{50} = 0.02$

Relative Frequency = $\frac{\text{Frequency}}{N}$
 $N = 50$



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1. X is wrong
 2. X is right :- A kind of curve
 3. X is right :- A kind of curve

if we have unequal width then we use
 additive frequency curve.

Numerical descriptions :-

Single not summary
 Pre requisite for an ideal measure

- Rigid definition
- easy to understand
- easy to calculate
- Based on all the observations
- suitable for further mathematical treatment
- least affected by sampling fluctuations & extreme values

central value $\rightarrow$ Measure of central Tendency
 spreadness $\rightarrow$ Measure of Dispersion

Measure of central tendency

1) Arithmetic Mean (Mean or Average)

$x_1, x_2, \dots, x_n \rightarrow n$ observations

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$= \frac{\sum_{i=1}^n x_i}{n}$$

for grouped data

$$\bar{x} = \frac{\sum f_i x_i}{N}$$

where $N = \sum f_i$

2) Median : middle most value of data when arranged in order

(i) arranged data in ascending order

(ii) Median = value of $(\frac{n+1}{2})^{th}$ observation

for grouped data

$$\text{Median} = L + \left[\frac{\left(\frac{n}{2} - cf \right)}{f} \right] \times h$$

L = lower limit of median class

n = less than cumulative frequency just preceding of class

h = width of median class

3) Mode : Most repeated value

Merely by inspection

For grouped data

$$\text{mode} = L + \left[\frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \right] \times h$$

L lower limit of modal class

f_1 = frequency of the modal class

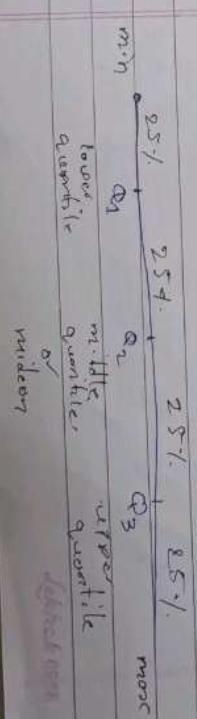
f_0 = frequency just preceding modal class

f_2 = frequency just succeeding modal class

h = width of modal class

Positional Averages :-

Quartiles :- Divide data into 4 equal parts.



$Q_i = \text{value of } i^{\text{th}} \left(\frac{n+1}{4} \right)^{\text{th}} \text{ observation}$

$i = 1, 2, 3$

for grouped data

$$Q_i = L_i + \frac{\left(\frac{n+1}{4} - m_i \right) \times h_i}{f_i}$$

$i = 1, 2, 3$

Measure of Dispersion :-

• Range = Max - min

• Variance : $(\sigma^2) = \frac{\sum (x_i - \bar{x})^2}{n}$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$= \frac{\sum x_i^2 - (\sum x_i)^2}{n}$$

$$= \frac{x_1^2 + x_2^2 + \dots + x_n^2}{n} - (\bar{x})^2$$

Standard deviation : (σ)

$$\sigma = \sqrt{\text{Variance}}$$

coefficient of variation

$$(C.V) = \frac{S.D}{\text{mean}} \times 100$$

1) calculate coeff

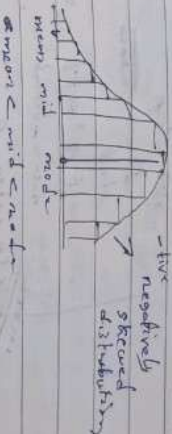
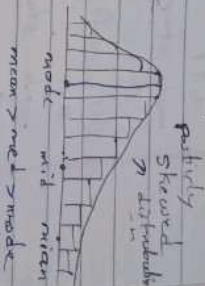
x (i) 4, 5, 6, 6, 7, 8, 9, 10, 12, 15

y (ii) 2, 3, 7, 6, 7, 8, 9, 20, 30, 40, 50

z (iii) 3, 5, 6, 6, 7, 8, 9, 10, 10, 12, 15, 15, 18

5 Measures

symmetry



graph is abstract in nature $\rightarrow$ distribution

coefficient of skewness

skewness method

$$Sk = \frac{\text{Mean} - \text{mode}}{S.D}$$

$$-3 \leq Sk \leq 3$$

$$S_b = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$-1 \leq S_b \leq 1$$

mode = most repeated value

for nominal $\rightarrow$ mode

for ordinal $\rightarrow$ midian

ratio $\rightarrow$ mean

ii) 4, 5, 6, 6, 6, 7, 8, 9, 10, 12, 15

1) mean = $\bar{x} = \frac{\sum x_i}{n}$